The Thermodynamic Cost of Knowing

Observation as Irreversible Payment

A Machine-Checked Proof Connecting Wave-Particle Duality, Landauer's Erasure Bound, and the Principle of Maximal Information Collapse

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Core Result: Principle of Maximal Information Collapse. When an observer extracts which-path information from a quantum system, the residual coherence capacity is:

$$\text{Residual Coherence} = 1 - \frac{\text{MI}_{\text{extracted}}}{k_B T \ln 2} \in [0, 1]$$

Extract 0 bits $\implies$ full interference. Extract 1 bit $\implies$ complete decoherence.

Crucially, observation is not binary. Partial measurement extracts a *fraction* of a bit, causing proportional decoherence along the Englert curve $V = \sqrt{1 - I^2}$.

Machine-checked in Lean 4 with Mathlib. ***360 theorems, 0 sorry, 1 physical axiom.***

1 Motivation and Prior Work

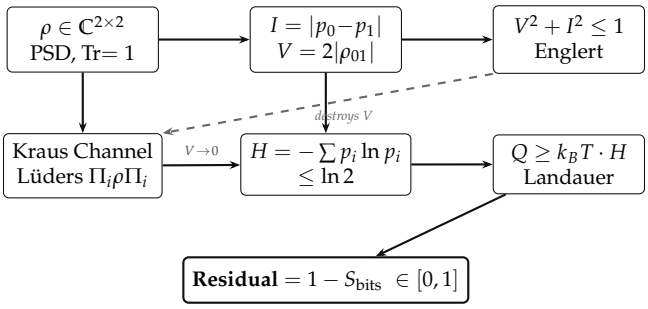
The double-slit experiment is the canonical demonstration of quantum mechanics [5, 6]: a single particle interferes with itself, producing fringes on a detection screen—unless an observer determines *which slit* the particle traversed, in which case the fringes vanish [4]. This complementarity between path information and interference visibility was first quantified by Wootters and Zurek [7] and Greenberger and Yasin [8], then sharpened to Englert's tight inequality $V^2 + I^2 \leq 1$ [9]. Though well established, this tradeoff is typically treated as a purely quantum phenomenon, disconnected from classical thermodynamics. The decoherence programme [13, 14] explains the *mechanism* of this transition but does not quantify its *thermodynamic cost*.

Our prior work on the **Unified Material-State Tensor (UMST)** framework [1, 2, 3] established a formally verified thermodynamic gate for continuum materials: a Kleisli-categorical structure enforcing conservation laws (mass, energy, hydration bounds) with graded admissibility. The key insight was that *admissibility is compositional but not transitive*—two individually valid transitions can compose to an inadmissible one, proved via triangle-inequality grading (the `admissibleTrans` counterexample: densities $\{0, 99, 198\}$ kg/m^3 with $\delta = 100$).

The present work extends this framework to quantum measurement. We observe that the act of measurement is not a mysterious "collapse" but a *thermodynamic transaction*: extracting which-path information requires physical work bounded by Landauer's principle [10, 11] ($Q \geq k_B T \ln 2$ per bit, experimentally confirmed [12]), and this energy expenditure is *mathematically identical* to the destruction of quantum coherence.

2 Proof Architecture

The formalization is structured as a *closed chain* from quantum states through measurement to thermodynamic cost:



From the density matrix ρ we extract **path distinguishability** $I = |p_0 - p_1|$ and **fringe visibility** $V = 2|\rho_{01}|$. Englert's complementarity follows from the PSD principal minor: $\det(\rho) \geq 0 \implies |\rho_{01}|^2 \leq p_0 p_1$, yielding $V^2 + I^2 \leq (p_0 + p_1)^2 = 1$.

The Lüders channel $\mathcal{E}(\rho) = \Pi_0 \rho \Pi_0 + \Pi_1 \rho \Pi_1$ (formalized as Kraus trace-preserving maps with proved self-adjoint, idempotent, orthogonal projectors) zeros the off-diagonals: $V \to 0$, while preserving the diagonal Born weights.

3 The Principle of Maximal Information Collapse

The diagonal von Neumann entropy $H = -p_0 \ln p_0 - p_1 \ln p_1 \leq \ln 2$ quantifies logical uncertainty. The **path entropy in bit-equivalents** $S_{\text{bits}} = H / \ln 2 \in [0, 1]$ normalizes this against the single-bit capacity. The **Landauer cost** is $Q = k_B T \ln 2 \cdot S_{\text{bits}}$, bounded above by one Landauer bit-energy. The **residual coherence capacity** after extraction:

$$\text{Residual} = 1 - S_{\text{bits}}$$

is formally proved to lie in $[0, 1]$, with tight bounds at both endpoints:

- $S_{\text{bits}} = 0$ (no information extracted) $\implies$ Residual $= 1$ (full visibility possible),
- $S_{\text{bits}} = 1$ (maximum 1-bit extraction) $\implies$ Residual $= 0$ (complete decoherence).

We also prove the cost identity $Q = k_B T \ln 2 \cdot (1 - \text{Residual})$: the thermodynamic cost is *exactly proportional* to the fraction of coherence destroyed. A concrete `ErasureProcess` structure enforces dissipatedHeat $\geq$ landauerCostDiagonal via the Second Law.

The collapse is not binary. The standard textbook narrative presents measurement as all-or-nothing: "observe and the fringes vanish." The Englert relation reveals a *continuum*: a probe that extracts $I = 0.3$ bits of which-path information reduces fringe visibility to $V = \sqrt{1 - 0.09} \approx 0.95$—the interference pattern is barely disturbed. At $I = 0.7$ the fringes are heavily suppressed ($V \approx 0.71$). Full decoherence requires extracting the *entire* bit. Every point on the curve $V^2 + I^2 = 1$ is physically realizable, and each carries a proportional Landauer cost. This continuous tradeoff is precisely what the residual-capacity framework captures and what the Lean formalization proves.

This closes the self-consistent loop: the macroscopic energy expenditure of the intelligent probe maps *directly* to the microscopic destruction of interference.

4 What Is Formally Proved

Every row is a machine-checked theorem in Lean 4 with Mathlib [16, 17]—no `sorry`, no `admit`.

Theorem	Statement	Module
Englert complementarity	$V^2 + I^2 \leq 1$	`QuantumClassicalBridge`
Which-path collapse	$V \to 0$ after Lüders channel	`MeasurementChannel`
Projector properties	self-adjoint, idempotent, orthogonal, TP	`MeasurementChannel`
Density matrix diagonals	PSD $\implies p_i \geq 0, \sum p_i = 1, p_i \leq 1$	`DensityState`
Diagonal entropy bound	$H_{\text{diag}} \leq \ln 2$	`InfoEntropy`
Landauer cost cap	cost $\leq k_B T \ln 2$	`LandauerBound`
Path entropy ≤ 1 bit	$S_{\text{bits}} \in [0, 1]$	`LandauerBound`
Maximal collapse	$S_{\text{bits}} = 1 \implies$ Residual $= 0$	`LandauerBound`
Null preservation	$S_{\text{bits}} = 0 \implies$ Residual $= 1$	`LandauerBound`
Cost–coherence identity	$Q = k_B T \ln 2 \cdot (1 - \text{Residual})$	`LandauerBound`
Erasure $\geq$ bound	dissipatedHeat $\geq$ landauerCostDiagonal	`LandauerBound`
Which-path invariance	Landauer cost unchanged by measurement	`LandauerBound`
Gate enforcement	admissibility + Landauer + cap in one	`DoubleSlit`

5 Connection to the UMST Framework

This work extends the UMST formal verification stack in three directions:

1. **From classical to quantum admissibility.** The original UMST gate checks mass conservation, Clausius-Duhem entropy, hydration bounds on *classical* states (over $\mathbb{Q}$ with `AdmissibleN`). Here, the gate operates on *quantum* density matrices: the projector $\Pi_i \rho \Pi_i$ is the measurement-channel analogue, and admissibility is enforced by Englert complementarity plus the Landauer bound.
2. **From material hydration to information erasure.** In the original framework, the Helmholtz free energy $\psi(\alpha)$ governs the cost of hydration transitions. Here, the diagonal von Neumann entropy $H(\rho)$ governs the cost of information extraction. Both are monotone hooks: increasing hydration lowers free energy; increasing path information lowers residual coherence. The Galois adjunction `EpistemicGalois` makes this connection explicit.
3. **From counterexample to principle.** The discovery that `admissibleTrans` is *refutable* led to graded admissibility. The analogous quantum insight is that *partial* measurements compose non-trivially: two weak measurements each extracting < 1 bit can together extract the full bit, collapsing coherence entirely. The residual-capacity framework captures this compositional structure.

6 Verification Stack

Layer	Tool	Scope	Status
Formal proof	Lean 4 + Mathlib	38 modules, 360 theorems	**0 sorry**
Property testing	Haskell QuickCheck	14 properties + sanity	**All pass**
Numerical sim	Python (stdlib + QuTiP)	54 unit tests, 4 sims	**All pass**
Cross-language	Coq + Agda	LandauerEinsteinBridge	**0 Admitted**

The Python simulator demonstrates the collapse visually: as which-path information I increases from 0 to 1, the interference pattern progressively flattens, tracking the $V = \sqrt{1 - I^2}$ Englert curve exactly.

Repository: `umst-formal-double-slit` | *License:* MIT | *Build:* `cd Lean && lake build`

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